

Qualification Pack



Cathodic Protection Technician

QP Code: HYC/Q6306

Version: 2.0

NSQF Level: 4

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Qualification Pack

Contents

HYC/Q6306: Cathodic Protection Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
HYC/N6312: Prepare cathodic protection system, tools and material for testing and maintenance activity	5
HYC/N6313: Carry out cathodic protection survey and testing for identifying damage in pipeline	9
HYC/N6314: Carry out repair and maintenance of pipelines using Cathodic Protection Techniques ...	14
HYC/N9301: Working Effectively in a team	19
HYC/N9302: Maintain health, safety and security procedures	23
DGT/VSQ/N0102: Employability Skills (60 Hours)	28
Assessment Guidelines and Weightage	35
<i>Assessment Guidelines</i>	35
<i>Assessment Weightage</i>	36
Acronyms	37
Glossary	38

Qualification Pack

HYC/Q6306: Cathodic Protection Technician

Brief Job Description

The cathodic protection technician is responsible for the prevention of metallic high-pressure hydrocarbon product pipeline/structure against the corrosion. The individual identifies the corrosion/damage in the pipeline and carry out the installation, commissioning and maintenance of the cathodic protection system. The person makes sure that the system is functioning correctly and providing adequate corrosion protection

Personal Attributes

The individual should have good sense of responsibility, a good organizer and time keeper. The person should have the ability to work independently.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N6312: Prepare cathodic protection system, tools and material for testing and maintenance activity](#)
2. [HYC/N6313: Carry out cathodic protection survey and testing for identifying damage in pipeline](#)
3. [HYC/N6314: Carry out repair and maintenance of pipelines using Cathodic Protection Techniques](#)
4. [HYC/N9301: Working Effectively in a team](#)
5. [HYC/N9302: Maintain health, safety and security procedures](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Midstream
Occupation	Operations - Oil and Gas pipeline, Operations - Oil and Gas pipeline
Country	India
NSQF Level	4

Qualification Pack

Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO/2015-NII
Minimum Educational Qualification & Experience	10th grade pass (with 2 years of relevant experience) OR 12th grade Pass OR 10th grade pass (plus 2-year of National Trade Certificate (NTC)) OR Completed 2nd year of the 3-year diploma after 10 (and pursuing regular Diploma)
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	17/11/2022
Version	2.0
Reference code on NQR	2022/HYC/HSSCI/06766
NQR Version	2.0

Remarks:

NA

Qualification Pack

HYC/N6312: Prepare cathodic protection system, tools and material for testing and maintenance activity

Description

This unit is about preparing for operational and maintenance activity required to be carried out before conducting cathodic protection

Scope

The scope covers the following :

- Prepare for operational and maintenance activities

Elements and Performance Criteria

Prepare for operational and maintenance activities

To be competent, the user/individual on the job must be able to:

- PC1.** obtain work permit to access the site location
- PC2.** assemble CP system and other necessary equipment required for testing and maintenance activity
- PC3.** check proper functioning of CP system and other equipment to ensure smooth operation
- PC4.** conduct adjustments in the CP system and other equipment, if any abnormalities have been found
- PC5.** compile and record the results of adjustments to optimise the integrity of equipment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principle of corrosion and its protection
- KU2.** the importance of calibration of tools and instruments
- KU3.** the basic principle of electrical and circuit system
- KU4.** the function of tools and instrument used for testing and maintenance
- KU5.** the importance of personal protective equipment
- KU6.** the Bureau of Indian Standards (BIS) 8062-2006 standards

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the data
- GS2.** write and prepare report in prescribed format
- GS3.** communicate effectively with team members

Qualification Pack

- GS4.** plan the work to meet the deadline
- GS5.** apply problem solving approaches in different situations
- GS6.** assess complexity and criticality of task

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for operational and maintenance activities</i>	20	30	-	-
PC1. obtain work permit to access the site location	3	4	-	-
PC2. assemble CP system and other necessary equipment required for testing and maintenance activity	5	8	-	-
PC3. check proper functioning of CP system and other equipment to ensure smooth operation	4	7	-	-
PC4. conduct adjustments in the CP system and other equipment, if any abnormalities have been found	3	4	-	-
PC5. compile and record the results of adjustments to optimise the integrity of equipment	5	7	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N6312
NOS Name	Prepare cathodic protection system, tools and material for testing and maintenance activity
Sector	Hydrocarbon
Sub-Sector	Midstream
Occupation	Operations - Oil and Gas pipeline, Operations - Oil and Gas pipeline
NSQF Level	4
Credits	3.5
Version	2.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

HYC/N6313: Carry out cathodic protection survey and testing for identifying damage in pipeline

Description

This unit is about identifying damages on pipeline through different CP System survey and testing methods.

Scope

The scope covers the following :

- Closed Interval Potential Logging (CIPL) Survey
- Coating integrity Survey
- Electrical isolation testing
- Detection/Testing of DC/AC Interference
- Electrical /electrolytic testing

Elements and Performance Criteria

Closed Interval Potential Logging (CIPL) Survey

To be competent, the user/individual on the job must be able to:

- PC1.** locate pipeline using pipeline locator
- PC2.** conduct survey at required interval along the pipeline as per the approved procedure
- PC3.** measure the polarized potential difference between the pipeline and an electrode on the ground
- PC4.** record the pipe to soil potential reading and compare it with the CP criterion to ensure that adequate levels of CP are reaching to the pipeline
- PC5.** determine the CIPL survey reading to identify the defects

Coating integrity Survey

To be competent, the user/individual on the job must be able to:

- PC6.** Identify the probable location of pipeline having defect based on the readings of Close Interval Potential Logging (CIPL) Survey
- PC7.** conduct Direct Current Voltage Gradient (DCVG) Survey at probable coating defect location
- PC8.** walk above pipeline with DCVG meter and two electrodes
- PC9.** measure the change in flow of electrical current through the ground
- PC10.** record the location where change in frequency is observed for further investigation

Electrical isolation testing

To be competent, the user/individual on the job must be able to:

- PC11.** perform visual inspection
- PC12.** measure the pipe-to-soil potential on both sides of the isolation device and confirm that the difference is adequate to assure positive isolation
- PC13.** check the continuity between both sides of the isolation device and confirm that the electric continuity does not exist

Qualification Pack

PC14. measure electrical insulation by clamp-on ammeter, continuous DC or pulsed DC holiday detector

PC15. ensure the joint/flange is insulated for minimum of 1 MΩ for new flange

PC16. report to maintenance supervisor in case short circuit is observed

Detection/Testing of DC/AC Interference

To be competent, the user/individual on the job must be able to:

PC17. check the interfering pipeline/metallic structure near cathodic protected buried pipeline

PC18. carry out close interval potential survey on affected pipeline by interrupting current source of protected pipeline (interfering pipeline)

PC19. record data by digital voltmeter in both ON and OFF state (only protected pipeline current source is interrupted)

PC20. measure the recorded data as per Bureau of Indian Standards (BIS) 8062

Electrical /electrolytic testing

To be competent, the user/individual on the job must be able to:

PC21. measure the flow of current from anode to the pipe using voltmeter

PC22. use galvanic series for predicting galvanic corrosion performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance of cathodic protection survey

KU2. the different types of survey method

KU3. the types of damages and corrosion found on oil and gas pipeline

KU4. the basic knowledge of earthing

KU5. the use of GIS system to identify pipeline network and GPS location based data recording for TLP

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret survey data and information correctly

GS2. communicate effectively with team members

GS3. plan the work to meet the deadline

GS4. apply problem solving approaches in different situations

GS5. assess complexity and criticality of task

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Closed Interval Potential Logging (CIPL) Survey</i>	6	17	-	-
PC1. locate pipeline using pipeline locator	1	3	-	-
PC2. conduct survey at required interval along the pipeline as per the approved procedure	2	4	-	-
PC3. measure the polarized potential difference between the pipeline and an electrode on the ground	1	3	-	-
PC4. record the pipe to soil potential reading and compare it with the CP criterion to ensure that adequate levels of CP are reaching to the pipeline	1	4	-	-
PC5. determine the CIPL survey reading to identify the defects	1	3	-	-
<i>Coating integrity Survey</i>	7	17	-	-
PC6. Identify the probable location of pipeline having defect based on the readings of Close Interval Potential Logging (CIPL) Survey	1	3	-	-
PC7. conduct Direct Current Voltage Gradient (DCVG) Survey at probable coating defect location	2	4	-	-
PC8. walk above pipeline with DCVG meter and two electrodes	1	3	-	-
PC9. measure the change in flow of electrical current through the ground	2	4	-	-
PC10. record the location where change in frequency is observed for further investigation	1	3	-	-
<i>Electrical isolation testing</i>	8	18	-	-
PC11. perform visual inspection	1	3	-	-
PC12. measure the pipe-to-soil potential on both sides of the isolation device and confirm that the difference is adequate to assure positive isolation	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. check the continuity between both sides of the isolation device and confirm that the electric continuity does not exist	2	3	-	-
PC14. measure electrical insulation by clamp-on ammeter, continuous DC or pulsed DC holiday detector	1	3	-	-
PC15. ensure the joint/flange is insulated for minimum of 1 MΩ for new flange	2	3	-	-
PC16. report to maintenance supervisor in case short circuit is observed	1	3	-	-
<i>Detection/Testing of DC/AC Interference</i>	6	12	-	-
PC17. check the interfering pipeline/metallic structure near cathodic protected buried pipeline	1	3	-	-
PC18. carry out close interval potential survey on affected pipeline by interrupting current source of protected pipeline (interfering pipeline)	2	3	-	-
PC19. record data by digital voltmeter in both ON and OFF state (only protected pipeline current source is interrupted)	2	3	-	-
PC20. measure the recorded data as per Bureau of Indian Standards (BIS) 8062	1	3	-	-
<i>Electrical /electrolytic testing</i>	3	6	-	-
PC21. measure the flow of current from anode to the pipe using voltmeter	1	3	-	-
PC22. use galvanic series for predicting galvanic corrosion performance	2	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N6313
NOS Name	Carry out cathodic protection survey and testing for identifying damage in pipeline
Sector	Hydrocarbon
Sub-Sector	Midstream
Occupation	Operations - Oil and Gas pipeline, Operations - Oil and Gas pipeline
NSQF Level	4
Credits	7.5
Version	2.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

HYC/N6314: Carry out repair and maintenance of pipelines using Cathodic Protection Techniques

Description

This unit is about carrying out repair and maintenance of pipeline through cathodic protection technique.

Scope

The scope covers the following :

- Identify suitable CP method
- Making cable connection from test station to pipeline using thermit weld/pin brazing
- Respond to problems
- Reporting

Elements and Performance Criteria

Identify suitable CP method

To be competent, the user/individual on the job must be able to:

- PC1.** review the findings on the basis of survey and testing to determine system abnormalities and maintenance requirement for the continued operation of the system
- PC2.** carry out suitable CP method from sacrificial anodic protection or impressed current cathodic protection based on the requirement
- PC3.** connect a highly anodic metal with the base metal to make base metal cathode under sacrificial anodic protection method
- PC4.** connect positive terminal of battery to backfill and negative terminal to the base metal to make base metal cathode under Impressed Current Cathodic Protection (ICCP) method
- PC5.** calculate the data for carrying out adjustments and maintenance of surface where abnormalities are found

Making cable connection from test station to pipeline using thermit weld/pin brazing

To be competent, the user/individual on the job must be able to:

- PC6.** connect cable from test station to pipe using thermit weld/pin brazing
- PC7.** remove the coating of pipe using abrasion tool
- PC8.** place copper cable on abraded pipe connected to test station
- PC9.** place graphite mould at the end of copper cable
- PC10.** insert thermit mixture powder into the mould
- PC11.** insert ignite powder into the mould
- PC12.** ignite the inserted powder using propane torch
- PC13.** remove mould and check the quality of connection by hammer blow
- PC14.** utilize the material required during repair and maintenance activity efficiently to minimize the waste and store the unused or reusable material

Respond to problems

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC15.** identify the possible problem during cathodic protection process or equipment
- PC16.** determine the possible cause of fault and action needed during maintenance of cathodic protection activity
- PC17.** apply suitable solution during maintenance of cathodic protection activity

Reporting

To be competent, the user/individual on the job must be able to:

- PC18.** maintain cathodic protection log book
- PC19.** report CP work completion to supervisor
- PC20.** report un-attended problem to the competent authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of cathodic protection survey
- KU2.** the different types of survey method
- KU3.** the types of cathodic protection
- KU4.** the types of damages and corrosion found on oil and gas pipeline
- KU5.** ways of efficiently utilizing the material for conducting maintenance of pipeline through cathodic protection technique

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret survey data and current for protecting metal from corrosion
- GS2.** write and prepare report in prescribed format
- GS3.** communicate effectively with team members
- GS4.** plan the work to meet the deadline
- GS5.** apply problem solving approaches in different situations
- GS6.** assess complexity and criticality of task

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify suitable CP method</i>	5	19	-	-
PC1. review the findings on the basis of survey and testing to determine system abnormalities and maintenance requirement for the continued operation of the system	1	4	-	-
PC2. carry out suitable CP method from sacrificial anodic protection or impressed current cathodic protection based on the requirement	1	4	-	-
PC3. connect a highly anodic metal with the base metal to make base metal cathode under sacrificial anodic protection method	1	4	-	-
PC4. connect positive terminal of battery to backfill and negative terminal to the base metal to make base metal cathode under Impressed Current Cathodic Protection (ICCP) method	1	4	-	-
PC5. calculate the data for carrying out adjustments and maintenance of surface where abnormalities are found	1	3	-	-
<i>Making cable connection from test station to pipeline using thermit weld/pin brazing</i>	15	32	-	-
PC6. connect cable from test station to pipe using thermit weld/pin brazing	2	4	-	-
PC7. remove the coating of pipe using abrasion tool	2	4	-	-
PC8. place copper cable on abraded pipe connected to test station	2	3	-	-
PC9. place graphite mould at the end of copper cable	2	4	-	-
PC10. insert thermit mixture powder into the mould	2	4	-	-
PC11. insert ignite powder into the mould	1	3	-	-
PC12. ignite the inserted powder using propane torch	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. remove mould and check the quality of connection by hammer blow	1	4	-	-
PC14. utilize the material required during repair and maintenance activity efficiently to minimize the waste and store the unused or reusable material	1	2	-	-
<i>Respond to problems</i>	6	10	-	-
PC15. identify the possible problem during cathodic protection process or equipment	2	4	-	-
PC16. determine the possible cause of fault and action needed during maintenance of cathodic protection activity	2	3	-	-
PC17. apply suitable solution during maintenance of cathodic protection activity	2	3	-	-
<i>Reporting</i>	4	9	-	-
PC18. maintain cathodic protection log book	2	3	-	-
PC19. report CP work completion to supervisor	1	3	-	-
PC20. report un-attended problem to the competent authority	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N6314
NOS Name	Carry out repair and maintenance of pipelines using Cathodic Protection Techniques
Sector	Hydrocarbon
Sub-Sector	Midstream
Occupation	Operations - Oil and Gas pipeline, Operations - Oil and Gas pipeline
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	NA
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

HYC/N9301: Working Effectively in a team

Description

This unit is about working effectively within a team.

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** provide support to the team members
- PC4.** respect the colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team
- PC8.** adopt gender neutral behaviour while interacting with colleagues
- PC9.** offer assistance to a person with disability (PWD), only if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** the importance of creating an environment of trust and mutual respect
- KU4.** the implications of own work on the work and schedule of others
- KU5.** the standard practices in organisation w.r.t communication at various levels
- KU6.** the personal responsibility for completing the task in time
- KU7.** importance of gender equality
- KU8.** importance of showing empathy while interacting with a PwD

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** work in a disciplined manner for meeting commitments and deadline
- GS4.** how to plan and prioritise the work
- GS5.** the importance of consistent and reliable services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	3	-	-
PC2. pass on information to colleagues in line with organisational requirements	2	3	-	-
PC3. provide support to the team members	2	4	-	-
PC4. respect the colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	2	3	-	-
PC6. inform team members timely, if timelines can't be met	2	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
PC8. adopt gender neutral behaviour while interacting with colleagues	2	2	-	-
PC9. offer assistance to a person with disability (PWD), only if required	2	3	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working Effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit is about maintaining health, safety and security procedure at workplace. It covers responsibilities towards self, others, assets and the environment.

Scope

The scope covers the following :

- Follow health and safety measures
- Follow safety procedures during emergency

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions
- PC2.** identify the people responsible for maintaining health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** follow safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** follow safety signages
- PC7.** maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub
- PC8.** inform the concerned person of any illness related to self and others
- PC9.** maintain workplace hygiene by disinfecting the equipment and tools regularly

Follow safety procedures during emergency

To be competent, the user/individual on the job must be able to:

- PC10.** respond promptly and appropriately to an accident or in an emergency situation
- PC11.** use appropriate fire extinguishers for different types of fires correctly
- PC12.** follow appropriate rescue techniques during fire hazard
- PC13.** follow good housekeeping practice in order to prevent fire hazards
- PC14.** inform fire safety department about any near-miss incidents in the work place
- PC15.** provide appropriate first aid to victims in an emergency situation
- PC16.** follow the applicable regulations and codes as per safety standard
- PC17.** prepare written accident/incident report and share with the concerned officer/department

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** health and safety hazards commonly affecting the work environment and related precautions
- KU4.** importance of maintaining personal hygiene using PPE kit, sanitizer and soap
- KU5.** importance of maintaining workplace hygiene
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** identify and report potential sources of danger
- GS4.** how to plan the work to meet the deadline
- GS5.** the importance of on time services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments in different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions	1	2	-	-
PC2. identify the people responsible for maintaining health and safety in the workplace	1	-	-	-
PC3. identify possible causes of risk or accident in the workplace	1	2	-	-
PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	-
PC6. follow safety signages	1	2	-	-
PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub	1	2	-	-
PC8. inform the concerned person of any illness related to self and others	1	1	-	-
PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly	1	2	-	-
<i>Follow safety procedures during emergency</i>	11	15	-	-
PC10. respond promptly and appropriately to an accident or in an emergency situation	1	2	-	-
PC11. use appropriate fire extinguishers for different types of fires correctly	2	2	-	-
PC12. follow appropriate rescue techniques during fire hazard	1	2	-	-
PC13. follow good housekeeping practice in order to prevent fire hazards	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC15. provide appropriate first aid to victims in an emergency situation	1	2	-	-
PC16. follow the applicable regulations and codes as per safety standard	1	2	-	-
PC17. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N6312.Prepare cathodic protection system, tools and material for testing and maintenance activity	20	30	0	0	50	15
HYC/N6313.Carry out cathodic protection survey and testing for identifying damage in pipeline	30	70	0	0	100	20
HYC/N6314.Carry out repair and maintenance of pipelines using Cathodic Protection Techniques	30	70	0	0	100	20
HYC/N9301.Working Effectively in a team	20	30	0	0	50	15
HYC/N9302.Maintain health, safety and security procedures	20	30	0	0	50	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	15
Total	140	260	0	0	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.